

BENCHMARK RESULTS REVIEW

Layout Detection Benchmarking Results Review

DocLayoutYOLO vs NVIDIA Nemotron-Parse-v1.1 vs LandingAI ADE-DPT2

● DocLayoutYOLO ● Nemotron-Parse-v1.1 ● ADE-DPT2 ● Ground Truth

Traditional public-dataset benchmark + Custom real-document benchmark vs human ground truth

Evaluation Setup

Models Evaluated

Model	Architecture	Params	Used in
DocLayoutYOLO	YOLO detector	~20M	Both benchmarks
Nemotron-Parse-v1.1	ViT-H encoder + mBart decoder	~885M	Both benchmarks
ADE-DPT2	Cloud API parser	N/A	Traditional only

Datasets / Documents Used

Dataset / Doc	Classes	Pages eval.	Notes
DocLayNet	11	3	COCO format
PubLayNet	5	3	COCO format
DocBank	9	3	Token-level → block aggregation
Custom medical catalog (30pg PDF)	7 present in GT	5 (human-annotated GT, 63 elements)	Real product document

Two Evaluation Tracks

Traditional benchmark — 3 models × 3 public datasets (DocLayNet, PubLayNet, DocBank), scored against published COCO-style annotations.

Custom benchmark — 2 models (DocLayoutYOLO, Nemotron-Parse) evaluated against human-labeled ground truth on a real 30-page medical product catalog.

Traditional Benchmark — Metrics

IoU (Intersection over Union)

$$IoU = \text{Area}(A \cap B) / \text{Area}(A \cup B)$$

Significance: Base unit of "how well two boxes overlap."

Threshold/Remark: IoU ≥ 0.50 is the standard match threshold.

mAP@50 / mAP@50:95

$$\begin{aligned} mAP@50:95 &= \text{avg } AP \text{ over } IoU \\ &0.50 \rightarrow 0.95 \text{ in steps of } 0.05, \\ &\text{then averaged across classes} \end{aligned}$$

Significance: mAP@50:95 is the stricter score (rewards tight boxes); mAP@50 is more forgiving.

Threshold/Remark: A big gap (e.g. 0.44 vs 0.30) means boxes are roughly right but not tightly localized.

Precision / Recall / F1

$$\begin{aligned} P &= TP / (TP + FP) & R &= TP / (TP + FN) \\ F1 &= 2PR / (P + R) \end{aligned}$$

Significance: P = share of predictions that were real; R = share of real items found; F1 = balance of both.

Threshold/Remark: F1 > 0.60 is generally considered usable.

Mean IoU

$$\begin{aligned} \text{Mean IoU} &= \text{avg IoU of matched} \\ &\text{(TP) pairs only} \end{aligned}$$

Significance: Measures localization tightness, ignoring misses entirely.

Threshold/Remark: Mean IoU > 0.85 indicates tight boxes.

11-pt Interpolated AP

$$\begin{aligned} AP &= (1/11) \cdot \sum \max\{P(r) : r \geq t\} \\ &\text{for } t = 0, 0.1, \dots, 1.0 \end{aligned}$$

Significance: Rewards a model that stays precise across all recall levels, not just one operating point.

Threshold/Remark: Closer to 1.0 = consistently strong across confidence thresholds.

Computed via pycocotools (COCO standard)

All five metrics above are reported per dataset, per model, on the following slide.

Traditional Benchmark — Results

Dataset	Model	mAP@50	mAP@50:95	Precision	Recall	F1	Mean IoU
DocLayNet	DocLayoutYOLO	0.436	0.352	0.588	0.556	0.571	0.894
DocLayNet	Nemotron-Parse	0.439	0.304	0.780	0.593	0.674	0.868
DocLayNet	ADE-DPT2	0.242	0.144	0.545	0.333	0.414	0.859
PubLayNet	DocLayoutYOLO	0.545	0.412	0.686	0.800	0.738	0.905
PubLayNet	Nemotron-Parse	0.383	0.224	0.818	0.600	0.692	0.884
PubLayNet	ADE-DPT2	0.490	0.275	0.739	0.567	0.642	0.848
DocBank	DocLayoutYOLO	0.016	0.007	0.233	0.184	0.206	0.761
DocBank	Nemotron-Parse	0.015	0.005	0.231	0.158	0.187	0.694
DocBank	ADE-DPT2	0.023	0.008	0.333	0.184	0.237	0.691

Note: DocBank collapses for all 3 models ($\text{mAP@50} \leq 0.023$) — likely an annotation-granularity mismatch, not true model failure; flagged for investigation, not a real performance gap.

Bold = best value per metric within each dataset group.

Custom Benchmark — Metrics (Layers 1-2)

Layer 1 — Geometric Match

*Hungarian-matched P / R / F1 / mean IoU
at IoU ≥ 0.50
Class Acc = correct-label matches /
total matched pairs*

Significance: Tests both "did we find the region" and "did we label it right" — separately.

Threshold/Remark: F1 > 0.60 good; Class Acc should track close to F1 — near 0 with healthy F1 signals a label-mapping bug, not real classification failure.

Layer 2 — COTe Trespass

*Trespass = Area(Pred outside GT)
/ Area(Pred)*

Significance: How much of the predicted area is "wasted" outside any true region.

Threshold/Remark: Trespass < 0.15 is good.

Layer 2 — COTe Coverage

*Coverage = Area(GT $\cap$ anyPred) /
Area(GT)*

Significance: Measures completeness — how much of the true region was found at all.

Threshold/Remark: Coverage > 0.70 is good.

Layer 2 — COTe Excess

Excess = Area(Pred) / Area(GT)

Significance: Over/under-detection ratio — total predicted area vs. total true area.

Threshold/Remark: Excess ≈ 1.0 is ideal (>1 over-detecting, <1 under-detecting).

Layer 2 — COTe Overlap

*Overlap = Area(GT $\cap$ doubly-covered)
/ Area(Pred)*

Significance: Captures redundant double-coverage of ground truth by predictions.

Threshold/Remark: Lower is better; should stay small relative to Coverage.

Pixel-mask based

Layer 2 metrics operate on rendered pixel masks rather than box coordinates, giving a finer-grained read on spatial quality than IoU alone.

Custom Benchmark — Metrics (Layers 3–4)

Layer 3 — LED Error Taxonomy

Thresholds: $IoU \geq 0.10$ "related" · ≥ 0.50 "matched" · ≥ 0.85 "duplicate"

Category	Definition	Category	Definition
Missing	GT with no related pred	Split	1 GT → multiple preds
Hallucination	Pred with no related GT	Merge	Multiple GT → 1 pred
Size-Error	Matched, $0.10 \leq IoU < 0.50$	Duplicate	2 preds, mutual $IoU \geq 0.85$
Misclassification	Matched, $IoU \geq 0.50$, wrong label	Overlap-Pred	2 preds, $0.10 \leq \text{mutual } IoU < 0.85$

Significance: Turns one aggregate error rate into "what kind of mistake is happening" — directly actionable. **Threshold/Remark:** Missing + Hallucination < 10% = healthy; Misclassification < 10% = good; anything dominating the breakdown points straight at the fix needed.

Layer 4 — Reading Order: ROKT

$$ROKT \text{ (Kendall's } \tau) = (C - D) / (n(n-1)/2)$$

on Hungarian-matched ($IoU \geq 0.30$)
GT vs predicted order

Significance: Tests whether the model reconstructs the correct reading sequence, not just the right boxes.

Threshold/Remark: $\tau > 0.70$ = strong agreement; 1.0 is perfect.

Layer 4 — Reading Order: ROA

$$ROA = 1 - (\text{adjacent inversions}) / (n - 1)$$

Significance: Measures local reading-order correctness via adjacent-pair inversions.

Threshold/Remark: 1.0 is perfect; lower values indicate more local order swaps.

Custom Benchmark — Results: Layer 1

Page	Model	Precision	Recall	F1	Mean IoU	Class Acc
7	DocLayoutYOLO	0.900	0.900	0.900	1.000	0.778
7	Nemotron-Parse	0.900	0.900	0.900	0.910	0.000
15	DocLayoutYOLO	1.000	0.952	0.976	0.997	0.350
15	Nemotron-Parse	0.944	0.810	0.872	0.771	0.000
17	DocLayoutYOLO	0.800	0.800	0.800	0.920	0.750
17	Nemotron-Parse	0.500	0.400	0.444	0.818	0.000
19	DocLayoutYOLO	0.250	0.077	0.118	0.833	0.000
19	Nemotron-Parse	0.846	0.846	0.846	0.723	0.000
25	DocLayoutYOLO	1.000	1.000	1.000	1.000	0.643
25	Nemotron-Parse	0.867	0.929	0.897	0.835	0.000
Mean	DocLayoutYOLO	0.790	0.746	0.759	0.950	0.504
Mean	Nemotron-Parse	0.811	0.777	0.792	0.811	0.000

Note: Nemotron's Class Accuracy is 0.000 on every page despite healthy F1 — almost certainly a label-taxonomy normalization bug, flagged for engineering follow-up, not read as a real result.

Custom Benchmark — Results: Layers 2–3

Layer 2 — Spatial Quality (COTe)

Page	Model	Coverage	Overlap	Trespass	Excess
7	DLY	0.952	0.000	0.000	0.952
7	Nemotron	0.988	0.000	0.049	1.039
15	DLY	0.940	0.000	0.000	0.940
15	Nemotron	0.165	0.000	0.182	0.202
17	DLY	0.948	0.000	0.011	0.959
17	Nemotron	0.503	0.000	0.037	0.522
19	DLY	0.928	0.000	0.396	1.535
19	Nemotron	0.874	0.000	0.046	0.916
25	DLY	1.000	0.000	0.000	1.000
25	Nemotron	0.855	0.000	0.291	1.206
Mean	DLY	0.954	0.000	0.081	1.077
Mean	Nemotron	0.677	0.000	0.121	0.777

Layer 3 — LED Error Taxonomy

Counts out of 53 DocLayoutYOLO / 60 Nemotron-Parse total detections

Error type	DocLayoutYOLO	Nemotron-Parse
Missing	11 (20.8%)	7 (11.7%)
Hallucination	3 (5.7%)	5 (8.3%)
Size-Error	2 (3.8%)	3 (5.0%)
Split	0 (0.0%)	0 (0.0%)
Merge	1 (1.9%)	1 (1.7%)
Overlap-Pred	1 (1.9%)	0 (0.0%)
Duplicate	0 (0.0%)	0 (0.0%)
Misclassification	22 (41.5%)	52 (86.7%)

Reading the two layers together: DocLayoutYOLO holds higher Coverage with lower Trespass on most pages, while Nemotron's Layer 3 breakdown is dominated by Misclassification (86.7%) — consistent with the Class Accuracy bug seen in Layer 1.

Layer 4 Results & Consolidated Scorecard

Layer 4 — Reading Order

Model	Kendall's τ	ROA
DocLayoutYOLO	1.000	1.000
Nemotron-Parse	1.000	1.000

Perfect reading-order agreement across all pages for both models.

Consolidated Scorecard

Layer	Metric	Threshold	DocLayoutYOLO	Nemotron-Parse
1	F1 @ IoU 0.50	>0.60	0.759	0.792
2	Coverage	>0.70	0.954	0.677
2	Trespass	<0.15	0.081	0.121
3	Missing + Hallucination	<10%	26.5%	20.0%
3	Misclassification	<10%	41.5%	86.7%
4	Kendall's τ	>0.70	1.000	1.000

Read the scorecard as: geometric localization and reading order are solid for both models; the breakdown is concentrated entirely in classification accuracy, most severely for Nemotron-Parse.

Key Findings & Next Steps

Key Findings

- Both models localize regions well and nail reading order — Layer 4 reading-order metrics hit a perfect 1.000 for both.
- Classification is the weak point for both models, and dramatically worse for Nemotron-Parse — likely a taxonomy-mapping bug that needs fixing before re-judging it.
- DocLayoutYOLO is far cheaper to run (CPU-capable, ~20M params) and processed all 30 document pages, vs. Nemotron's 5.
- Sample sizes in both benchmarks are small (3 pages/dataset, 5 GT pages) — treat results as directional, not definitive.

Next Steps

- Fix the Nemotron-Parse & DocLayoutYOLO label-taxonomy mapping bug, and map all the labels pred and GT in an equivalent manner , this will clarify some evaluation metric errors also.
- Re-run both benchmarks with larger samples for statistical confidence.
- Add ADE-DPT2 to the custom benchmark for full 3-way parity.

Models Referenced

- DocLayoutYOLO — fast, cheap, CPU-capable
- Nemotron-Parse-v1.1 — strong but mislabeled classes
- ADE-DPT2 — traditional benchmark only, to date